

Kiyas Mahmud

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PROFILE

Computer Science undergraduate specializing in machine learning, full-stack development, and AI systems. Built Hybrid Fraud Shield, an award-winning 13-model ensemble fraud detection system recognized at APICTA 2025, and won WICE 2026 for ScholarNest, an AI-powered full-stack SaaS application. Experienced in deploying scalable, production-ready software across ML and web platforms. Driven by building practical AI solutions that solve real-world problems.

EDUCATION

American International University-Bangladesh

BSc in Computer Science

Dhaka, Bangladesh

Expected 2026

Ideal College, Dhanmondi

Higher Secondary Certificate, Science

Dhaka, Bangladesh

2021

Singair Pilot Government High School

Secondary School Certificate, Science

Manikganj, Bangladesh

2018

EXPERIENCE

Tech Wing's Lab

Research Assistant

Dhaka, Bangladesh

January 2025 – March 2026

- Support research design, analysis, documentation, and publication preparation.
- Mentor junior researchers in academic writing and research methodology.
- Review manuscripts for clarity, structure, formatting, and technical accuracy.
- Assist with citation checks and paper refinement for publication readiness.

PROJECTS

ScholarNest

AI-Powered Global Scholarship Command Center 

- Built a scholarship tracker with global discovery, smart filters, deadline alerts, application pipelines, and document checklists.
- Developed professor CRM features for AI-personalized emails, follow-up scheduling, reply tracking, and smart notifications.
- Added document vault, analytics dashboard, risk scoring, AI document review, and Telegram/email reminder workflows.

Tools: Nuxt 3, Vue.js, TypeScript, PostgreSQL, Prisma ORM, Tailwind CSS, Redis, Cloudflare R2, Gemini AI

Hybrid Fraud Shield

AI-Based Real-Time Fraud Detection

- Built a hybrid fraud detection system using XGBoost, CatBoost, CNN, and BiLSTM models.
- Applied meta-learning fusion to combine machine learning and deep learning predictions.
- Cleaned, verified, and validated labeled transaction datasets for model training.
- Added Explainable AI support using SHAP and Integrated Gradients.
- Deployed real-time risk analysis using FastAPI.

Tools: Python, Jupyter Notebook, Scikit-learn, XGBoost, CatBoost, SHAP, FastAPI

UniFind

Centralized Campus Lost & Found Management Platform

- Built a campus lost-and-found portal with global item discovery, multi-image posting, advanced filtering by location, category, and status, and verification-based claim pipelines.
- Developed admin moderation workflows for evidence review, dispute resolution, post lifecycle tracking, user reporting, and dynamic master data management.
- Added Role-Based Access Control, modular RESTful APIs, an interactive analytics dashboard, session security, and structured MySQL database migrations.

Tools: PHP 8, MySQL, JavaScript (ES6+), HTML5, CSS3, REST API, Apache, Laragon, Git

ResearchLink

Academic Research and Recruitment Desktop App

- Developed a desktop application for managing research papers and job postings.
- Implemented role-based access for admins, authors, and researchers.
- Enabled paper submission, job posting, applications, and status tracking.

Tools: C#, .NET WinForms, SQL Server

Things You Desire

E-Commerce Clothing Platform

- Built customer and admin applications for an e-commerce clothing platform.
- Implemented guest checkout, product variants, and inventory synchronization.
- Added promotional banners, click analytics, and sales reporting.

Tools: Laravel, Next.js, MySQL, Tailwind CSS, ShadCN UI, Framer Motion, Laravel Sanctum

TECHNICAL SKILLS

Languages C++, JavaScript, TypeScript, PHP, Python

Frameworks Laravel, React, Next.js, Vue.js, Nuxt, Scikit-learn

Data/ML Pandas, NumPy, Jupyter Notebook, XGBoost, CatBoost, SHAP, dataset cleaning, label validation

Core Areas Data Structures, Algorithms, OOP, Database Management, Data Science, Machine Learning

PUBLICATIONS

AutoTuneX: An Intelligent Self-Optimizing Runtime Framework for Cloud Applications

Accepted at IEEE TENCON 2026, Bali, Indonesia

- Designed a five-component closed-loop Kubernetes autoscaling framework: Monitor, Feature Engineer, Predictor, Decision Engine, and Safety Validator.
- Achieved 72.2% cost reduction with zero SLO violations and zero oscillations using a lightweight Linear Regression predictor, validated through a live Minikube deployment.

Hybrid FraudShield: Detecting Financial Fraud in Real Time with Explainable AI

Accepted at ICECCME 2026, Bali, Indonesia

- Built an 11-model stacking ensemble using five machine learning and six deep learning models with a Logistic Regression meta-learner on the IEEE-CIS dataset of 590,000 transactions.
- Achieved 98.7% accuracy, 0.978 ROC-AUC, and an 8.1 percentage-point precision improvement over XGBoost at the same recall.
- Built a serving pipeline with real-time SHAP explanations under 500 ms and a rule-based velocity-attack detection layer.

Comparative Analysis of Heap Algorithms for Anomaly Localization in Dynamic Graph Neural Networks

Published in Atlantis Press, Proceedings of ICERIE 2025; available online 18 November 2025

- Evaluated Fibonacci, D-ary, and PriorityQueue heaps within Dijkstra's Algorithm.
- Focused on speed, memory efficiency, and scalability for anomaly localization.

Real-Time Fusion of Behavioral and Transactional Data for Fraud Detection

Presented at ICASF 2025, Dubai

- Proposed a real-time machine learning framework for financial fraud detection.
- Integrated behavioral, transactional, and device data using ensemble learning.

COURSES AND CERTIFICATES

Machine Learning with Python – IBM, Coursera

- Covered regression, classification, clustering, and model evaluation using Python.

PHP and Laravel, Batch 2 – OSTAD

- Completed backend development training in Laravel, databases, APIs, and deployment.

ACHIEVEMENTS

Silver Medal – World Invention Competition & Exhibition Bangladesh, WICEBD

- Awarded for *ScholarNest*, a research collaboration platform supporting academic discovery and researcher networking.

Winner – Bangladesh ICT & Innovation Awards 2025

- Recognized nationally for *Hybrid Fraud Shield*, an AI-based real-time fraud detection framework.

Participant – Asia Pacific ICT Alliance Awards 2025

- Represented *Hybrid Fraud Shield* in a regional ICT innovation competition.

LANGUAGES

English: Fluent, professional working proficiency

Bengali/Bangla: Native